

```

> restart: unprotect('D'):

m := 3;

twist := - 2;

d := 8;                degree := d + 1:

##

SS := x^(d+1) + 1^(d+1) + 0*x^6:
HSS := X^(d+1) + T^(d+1) + 0*X^6:

#SS := x^(d+1) + 1^(d+1) + 1*x^6:
#HSS := X^(d+1) + T^(d+1) + 1*T^(d+1-6)*X^6:

##

Jet_xR := expand(
+ (1/Ry^(m-0*2)) * add(AA||j * x1^(m-0*3-j) * (R1/RR)^j *
Delta_xR^0, j=0..m-0*3)
+ (1/Ry^(m-1*2)) * add(BB||j * x1^(m-1*3-j) * (R1/RR)^j *
Delta_xR^1, j=0..m-1*3)
+ (1/Ry^(m-2*2)) * add(CC||j * x1^(m-2*3-j) * (R1/RR)^j *
Delta_xR^2, j=0..m-2*3)
+ (1/Ry^(m-3*2)) * add(DD||j * x1^(m-3*3-j) * (R1/RR)^j *
Delta_xR^3, j=0..m-3*3)
+ (1/Ry^(m-4*2)) * add(EE||j * x1^(m-4*3-j) * (R1/RR)^j *
Delta_xR^4, j=0..m-4*3) ):

##

for l from 0 to m - 0*3 do dA[l] := (m-0*2)*d - m + l*1 + 0*1 +
twist: od:

for l from 0 to m - 1*3 do dB[l] := (m-1*2)*d - m + l*1 + 1*1 +
twist: od:

for l from 0 to m - 2*3 do dC[l] := (m-2*2)*d - m + l*1 + 2*1 +
twist: od:

for l from 0 to m - 3*3 do dD[l] := (m-3*2)*d - m + l*1 + 3*1 +
twist: od:

for l from 0 to m - 4*3 do dE[l] := (m-4*2)*d - m + l*1 + 4*1 +
twist: od:

##

```

```

Delta_xR := x1 * (R2/RR - R1*R1/RR^2) - x2 * (R1/RR):
Solx1 := - y1*Ry/Rx + R1/Rx:
Solx2 := - Rxx*y1^2*Ry^2/Rx^3 + 2*Rxx*y1*Ry*R1/Rx^3 - Rxx*
R1^2/Rx^3 + 2*y1^2*Rxy*Ry/Rx^2
          - 2*y1*Rxy*R1/Rx^2 - y1^2*Ryy/Rx - y2*Ry/Rx + R2/Rx:
##
Jet_Ry := sort(mtaylor(
          expand(subs(R2=-(RR^2*Delta_Ry-RR*R1*y2-
R1^2*y1)/(RR*y1),
          expand(subs({x1=Solx1,x2=Solx2},
Jet_xR))),
          [R1,y1,Delta_Ry], 100), [R1,y1,Delta_Ry]):
##
printlevel := 2:
for k from 0 to m/3 do
for j from 0 to m-3*k do
EEq[m-j-3*k,j,k] := factor(
  Ry^(m-j-3*k)*RR^(m-j-3*k)*Rx^(m-2*k)
*
  coeftayl(Jet_Ry, [R1,y1,Delta_Ry]=[0,0,0], [m-j-3*k,j,k]) ):
od: od:
## EQUATION RR EXPLICITE
RR := y^(d+1) + SS;
HRR := Y^(d+1) + HSS;
RY := diff(HRR, Y):
Rx := diff(RR,x):
Ry := diff(RR,y):
Rxx := diff(RR,x,x):
Rxy := diff(RR,x,y):
Ryy := diff(RR,y,y):
##
printlevel := 2:
for k from 0 to m/3 do
for j from 0 to m-3*k do
Eq[m-j-3*k,j,k] := mtaylor(EEq[m-j-3*k,j,k], y, (m-j-3*k)*d):
Var[m-j-3*k,j,k] := indets(Eq[m-j-3*k,j,k]) minus {x,y}:

```

od: od:

##

$m := 3$

$twist := -2$

$d := 8$

$RR := x^9 + y^9 + 1$

$HRR := T^9 + X^9 + Y^9$

(1)

> ## EXPAND ALL UNKNOWNNS UP TO $y^{(1*d)}$

for l from 0 to m - 0*3 do
AA[l] := add(add(A[l][i,j]*x^i*y^j, i=0..dA[l]-j), j=0..min(dA[l], 1*d-1)): od:

for l from 0 to m - 1*3 do
BB[l] := add(add(B[l][i,j]*x^i*y^j, i=0..dB[l]-j), j=0..min(dB[l], 1*d-1)): od:

for l from 0 to m - 2*3 do
CC[l] := add(add(C[l][i,j]*x^i*y^j, i=0..dC[l]-j), j=0..min(dC[l], 1*d-1)): od:

for l from 0 to m - 3*3 do
DD[l] := add(add(D[l][i,j]*x^i*y^j, i=0..dD[l]-j), j=0..min(dD[l], 1*d-1)): od:

for l from 0 to m - 4*3 do
EE[l] := add(add(E[l][i,j]*x^i*y^j, i=0..dE[l]-j), j=0..min(dE[l], 1*d-1)): od:

SYSTEM FOR DIVISIBILITY BY $y^{(1*d)}$

Systemey1d := {}:

for k from 0 to (m-1)/3 do

Coeffs_y1d[1,m-1-3*k,k] := {coeffs(expand(mtaylor(Eq[1,m-1-3*k,k], y, 1*d)), [x,y])}: od:

Systemey1d := Systemey1d union Coeffs_y1d[1,m-1-3*k,k]:

od:

Systeme_y1d := Systemey1d:

VARIABLES FOR DIVISIBILITY BY $y^{(1*d)}$

Variables_y1d := indets(Systeme_y1d):

```

## RESOLUTION  $y^{(1*d)}$ 

Solutions_y1d := solve(Systeme_y1d, Variables_y1d):

assign(Solutions_y1d):

##

for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
for l from 0 to m - 4*3 do nops(expand(EE||l)) od;

```

$l := 0$

68

$l := 1$

137

$l := 2$

148

$l := 3$

156

$l := 4$

$l := 0$

19

$l := 1$

$l := 0$

$l := 0$

$l := 0$

(2)

> ## EXPAND ALL UNKNOWNNS UP TO $y^{(2*d)}$

```

for l from 0 to m - 0*3 do
AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j), j=0..min(dA[l],2*d-1)): od:

```

```

for l from 0 to m - 1*3 do
BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j), j=0..min(dB[l],2*d-1)): od:

```

```

for l from 0 to m - 2*3 do
CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j), j=0..min(dC[l],2*d-1)): od:

```

```

for l from 0 to m - 3*3 do
DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j), j=0..min(dD
[l],2*d-1)): od:

for l from 0 to m - 4*3 do
EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j), j=0..min(dE
[l],2*d-1)): od:

## SYSTEM FOR DIVISIBILITY BY y^(2*d)

Systemey2d := {}:

for k from 0 to (m-2)/3 do

Coeffs_y2d[2,m-2-3*k,k] := {coeffs(expand(mtaylor(Eq[2,m-2-3*k,
k], y, 2*d)), [x,y])}:

Systemey2d := Systemey2d union Coeffs_y2d[2,m-2-3*k,k]:

od:

Systeme_y2d := Systemey2d:

## VARIABLES FOR DIVISIBILITY BY y^(2*d)

Variables_y2d := indets(Systeme_y2d):

## RESOLUTION y^(2*d)

Solutions_y2d := solve(Systeme_y2d, Variables_y2d):

assign(Solutions_y2d):

##

for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
for l from 0 to m - 4*3 do nops(expand(EE||l)) od;

l := 0
20
l := 1
50
l := 2
90
l := 3
248
l := 4
l := 0

```

15

$l := 1$

$l := 0$

$l := 0$

$l := 0$

(3)

```
> ## EXPAND ALL UNKNOWNNS UP TO  $y^{(3*d)}$ 

for l from 0 to m - 0*3 do
AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j), j=0..min(dA
[l],3*d-1)): od:

for l from 0 to m - 1*3 do
BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j), j=0..min(dB
[l],3*d-1)): od:

for l from 0 to m - 2*3 do
CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j), j=0..min(dC
[l],3*d-1)): od:

for l from 0 to m - 3*3 do
DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j), j=0..min(dD
[l],3*d-1)): od:

for l from 0 to m - 4*3 do
EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j), j=0..min(dE
[l],3*d-1)): od:

## SYSTEM FOR DIVISIBILITY BY  $y^{(3*d)}$ 

Systemey3d := {}:

for k from 0 to (m-3)/3 do

Coeffs_y3d[3,m-3-3*k,k] := {coeffs(expand(mtaylor(Eq[3,m-3-3*k,
k], y, 3*d)), [x,y])}:

Systemey3d := Systemey3d union Coeffs_y3d[3,m-3-3*k,k]:

od:

Systeme_y3d := Systemey3d:

## VARIABLES FOR DIVISIBILITY BY  $y^{(3*d)}$ 

Variables_y3d := indets(Systeme_y3d):

## RESOLUTION  $y^{(3*d)}$ 

Solutions_y3d := solve(Systeme_y3d, Variables_y3d):
```

```
assign(Solutions_y3d):
```

```
##
```

```
for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
for l from 0 to m - 4*3 do nops(expand(EE||l)) od;
```

```
l := 0
```

```
1
```

```
l := 1
```

```
6
```

```
l := 2
```

```
6
```

```
l := 3
```

```
1
```

```
l := 4
```

```
l := 0
```

```
6
```

```
l := 1
```

```
l := 0
```

```
l := 0
```

```
l := 0
```

(4)

```
> ## EXPAND ALL UNKNOWNNS UP TO y^(4*d)
```

```
for l from 0 to m - 0*3 do
AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j), j=0..min(dA
[l],4*d-1)): od:
```

```
for l from 0 to m - 1*3 do
BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j), j=0..min(dB
[l],4*d-1)): od:
```

```
for l from 0 to m - 2*3 do
CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j), j=0..min(dC
[l],4*d-1)): od:
```

```
for l from 0 to m - 3*3 do
DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j), j=0..min(dD
[l],4*d-1)): od:
```

```
for l from 0 to m - 4*3 do
```

```

EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j), j=0..min(dE
[l],4*d-1)): od:

## SYSTEM FOR DIVISIBILITY BY y^(4*d)

Systemey4d := {}:

for k from 0 to (m-4)/3 do

Coeffs_y4d[4,m-4-3*k,k] := {coeffs(expand(mtaylor(Eq[4,m-4-3*k,
k], y, 4*d)), [x,y])}:

Systemey4d := Systemey4d union Coeffs_y4d[4,m-4-3*k,k]:

od:

Systeme_y4d := Systemey4d:

## VARIABLES FOR DIVISIBILITY BY y^(4*d)

Variables_y4d := indets(Systeme_y4d):

## RESOLUTION y^(4*d)

Solutions_y4d := solve(Systeme_y4d, Variables_y4d):

assign(Solutions_y4d):

##

for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
for l from 0 to m - 4*3 do nops(expand(EE||l)) od;

l := 0
1
l := 1
6
l := 2
6
l := 3
1
l := 4
l := 0
6
l := 1
l := 0
l := 0

```



```

> ## EXPAND ALL UNKNOWNNS UP TO  $y^{(5*d)}$ 

for l from 0 to m - 0*3 do
AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j), j=0..min(dA[l],5*d-1)): od:

for l from 0 to m - 1*3 do
BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j), j=0..min(dB[l],5*d-1)): od:

for l from 0 to m - 2*3 do
CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j), j=0..min(dC[l],5*d-1)): od:

for l from 0 to m - 3*3 do
DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j), j=0..min(dD[l],5*d-1)): od:

for l from 0 to m - 4*3 do
EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j), j=0..min(dE[l],5*d-1)): od:

## SYSTEM FOR DIVISIBILITY BY  $y^{(5*d)}$ 

Systemey5d := {}:

for k from 0 to (m-5)/3 do

Coeffs_y5d[5,m-5-3*k,k] := {coeffs(expand(mtaylor(Eq[5,m-5-3*k,
k], y, 5*d)), [x,y])}:

Systemey5d := Systemey5d union Coeffs_y5d[5,m-5-3*k,k]:

od:

Systeme_y5d := Systemey5d:

## VARIABLES FOR DIVISIBILITY BY  $y^{(5*d)}$ 

Variables_y5d := indets(Systeme_y5d):

## RESOLUTION  $y^{(5*d)}$ 

Solutions_y5d := solve(Systeme_y5d, Variables_y5d):

assign(Solutions_y5d):

##

for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;

```

```

for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
for l from 0 to m - 4*3 do nops(expand(EE||l)) od;

```

```
l := 0
```

```
1
```

```
l := 1
```

```
6
```

```
l := 2
```

```
6
```

```
l := 3
```

```
1
```

```
l := 4
```

```
l := 0
```

```
6
```

```
l := 1
```

```
l := 0
```

```
l := 0
```

```
l := 0
```

(6)

```
> ## EXPAND ALL UNKNOWNNS UP TO  $y^{(6*d)}$ 
```

```

for l from 0 to m - 0*3 do
AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j), j=0..min(dA
[l],6*d-1)): od:

```

```

for l from 0 to m - 1*3 do
BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j), j=0..min(dB
[l],6*d-1)): od:

```

```

for l from 0 to m - 2*3 do
CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j), j=0..min(dC
[l],6*d-1)): od:

```

```

for l from 0 to m - 3*3 do
DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j), j=0..min(dD
[l],6*d-1)): od:

```

```

for l from 0 to m - 4*3 do
EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j), j=0..min(dE
[l],6*d-1)): od:

```

```
## SYSTEM FOR DIVISIBILITY BY  $y^{(6*d)}$ 
```

```

Systemey6d := {}:
for k from 0 to (m-6)/3 do
  Coeffs_y6d[6,m-6-3*k,k] := {coeffs(expand(mtaylor(Eq[6,m-6-3*k,
k], y, 6*d)), [x,y])}:
Systemey6d := Systemey6d union Coeffs_y6d[6,m-6-3*k,k]:
od:
Systeme_y6d := Systemey6d:
## VARIABLES FOR DIVISIBILITY BY y^(6*d)
Variables_y6d := indets(Systeme_y6d):
## RESOLUTION y^(6*d)
Solutions_y6d := solve(Systeme_y6d, Variables_y6d):
assign(Solutions_y6d):
##
for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
for l from 0 to m - 4*3 do nops(expand(EE||l)) od;

l := 0
1
l := 1
6
l := 2
6
l := 3
1
l := 4
l := 0
6
l := 1
l := 0
l := 0
l := 0

```

```

> ## EXPAND ALL UNKNOWNNS UP TO  $y^{(7*d)}$ 

for l from 0 to m - 0*3 do
AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j), j=0..min(dA
[l], 7*d-1)): od:

for l from 0 to m - 1*3 do
BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j), j=0..min(dB
[l], 7*d-1)): od:

for l from 0 to m - 2*3 do
CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j), j=0..min(dC
[l], 7*d-1)): od:

for l from 0 to m - 3*3 do
DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j), j=0..min(dD
[l], 7*d-1)): od:

for l from 0 to m - 4*3 do
EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j), j=0..min(dE
[l], 7*d-1)): od:

## SYSTEM FOR DIVISIBILITY BY  $y^{(7*d)}$ 

Systemey7d := {}:

for k from 0 to (m-7)/3 do

Coeffs_y7d[7,m-7-3*k,k] := {coeffs(expand(mtaylor(Eq[7,m-7-3*k,
k], y, 7*d)), [x,y])}:

Systemey7d := Systemey7d union Coeffs_y7d[7,m-7-3*k,k]:

od:

Systeme_y7d := Systemey7d:

## VARIABLES FOR DIVISIBILITY BY  $y^{(7*d)}$ 

Variables_y7d := indets(Systeme_y7d):

## RESOLUTION  $y^{(7*d)}$ 

Solutions_y7d := solve(Systeme_y7d, Variables_y7d):

assign(Solutions_y7d):

##

for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
for l from 0 to m - 4*3 do nops(expand(EE||l)) od;

```

$l := 0$

```

1
l := 1
6
l := 2
6
l := 3
1
l := 4
l := 0
6
l := 1
l := 0
l := 0
l := 0

```

(8)

```

>
## EXPAND ALL UNKNOWNNS UP TO  $y^{(8*d)}$ 

for l from 0 to m - 0*3 do
AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j), j=0..min(dA[l],8*d-1)): od:

for l from 0 to m - 1*3 do
BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j), j=0..min(dB[l],8*d-1)): od:

for l from 0 to m - 2*3 do
CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j), j=0..min(dC[l],8*d-1)): od:

for l from 0 to m - 3*3 do
DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j), j=0..min(dD[l],8*d-1)): od:

for l from 0 to m - 4*3 do
EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j), j=0..min(dE[l],8*d-1)): od:

## SYSTEM FOR DIVISIBILITY BY  $y^{(8*d)}$ 

Systemey8d := {}:

for k from 0 to (m-8)/3 do

Coeffs_y8d[8,m-8-3*k,k] := {coeffs(expand(mtaylor(Eq[8,m-8-3*k,
k], y, 8*d)), [x,y])}:

```

```

Systemey8d := Systemey8d union Coeffs_y8d[8,m-8-3*k,k]:
od:
Systeme_y8d := Systemey8d:
## VARIABLES FOR DIVISIBILITY BY  $y^{(8*d)}$ 
Variables_y8d := indets(Systeme_y8d):
## RESOLUTION  $y^{(8*d)}$ 
Solutions_y8d := solve(Systeme_y8d, Variables_y8d):
assign(Solutions_y8d):
##
for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
for l from 0 to m - 4*3 do nops(expand(EE||l)) od;

l := 0
1
l := 1
6
l := 2
6
l := 3
1
l := 4
l := 0
6
l := 1
l := 0
l := 0
l := 0

```

(9)

```

> ## EXPAND ALL UNKNOWNNS UP TO  $y^{(9*d)}$ 
for l from 0 to m - 0*3 do
AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j), j=0..min(dA
[l],9*d-1)): od:

```

```

for l from 0 to m - 1*3 do
BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j), j=0..min(dB
[l],9*d-1)): od:

for l from 0 to m - 2*3 do
CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j), j=0..min(dC
[l],9*d-1)): od:

for l from 0 to m - 3*3 do
DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j), j=0..min(dD
[l],9*d-1)): od:

for l from 0 to m - 4*3 do
EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j), j=0..min(dE
[l],9*d-1)): od:

## SYSTEM FOR DIVISIBILITY BY y^(9*d)

Systemey9d := {}:

for k from 0 to (m-9)/3 do

Coeffs_y9d[9,m-9-3*k,k] := {coeffs(expand(mTaylor(Eq[9,m-9-3*k,
k], y, 9*d)), [x,y])}:

Systemey9d := Systemey9d union Coeffs_y9d[9,m-9-3*k,k]:

od:

Systeme_y9d := Systemey9d:

## VARIABLES FOR DIVISIBILITY BY y^(9*d)

Variables_y9d := indets(Systeme_y9d):

## RESOLUTION y^(9*d)

Solutions_y9d := solve(Systeme_y9d, Variables_y9d):

assign(Solutions_y9d):

##

for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
for l from 0 to m - 4*3 do nops(expand(EE||l)) od;

l := 0
1
l := 1
6
l := 2

```

```

6
l := 3
1
l := 4
l := 0
6
l := 1
l := 0
l := 0
l := 0

```

(10)

```

>
## EXPAND ALL UNKNOWNNS UP TO y^(10*d)

for l from 0 to m - 0*3 do
AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j), j=0..min(dA
[l],10*d-1)): od:

for l from 0 to m - 1*3 do
BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j), j=0..min(dB
[l],10*d-1)): od:

for l from 0 to m - 2*3 do
CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j), j=0..min(dC
[l],10*d-1)): od:

for l from 0 to m - 3*3 do
DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j), j=0..min(dD
[l],10*d-1)): od:

for l from 0 to m - 4*3 do
EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j), j=0..min(dE
[l],10*d-1)): od:

## SYSTEM FOR DIVISIBILITY BY y^(10*d)

Systemey10d := {}:

for k from 0 to (m-10)/3 do

Coeffs_y10d[10,m-10-3*k,k] := {coeffs(expand(mtaylor(Eq[10,m-10
-3*k,k], y, 10*d)), [x,y])}:

Systemey10d := Systemey10d union Coeffs_y10d[10,m-10-3*k,k]:

od:

Systeme_y10d := Systemey10d:

```



```

## VARIABLES FOR DIVISIBILITY BY  $y^{(10*d)}$ 
Variables_y10d := indets(Systeme_y10d):
## RESOLUTION  $y^{(10*d)}$ 
Solutions_y10d := solve(Systeme_y10d, Variables_y10d):
assign(Solutions_y10d):
##
for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
for l from 0 to m - 4*3 do nops(expand(EE||l)) od;

l := 0
1
l := 1
6
l := 2
6
l := 3
1
l := 4
l := 0
6
l := 1
l := 0
l := 0
l := 0

```

(11)

```

> ## EXPAND ALL UNKNOWNNS UP TO  $y^{(11*d)}$ 
for l from 0 to m - 0*3 do
AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j), j=0..min(dA[l],11*d-1)): od;

for l from 0 to m - 1*3 do
BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j), j=0..min(dB[l],11*d-1)): od;

for l from 0 to m - 2*3 do

```

```

CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j), j=0..min(dC
[l],11*d-1)): od:

for l from 0 to m - 3*3 do
DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j), j=0..min(dD
[l],11*d-1)): od:

for l from 0 to m - 4*3 do
EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j), j=0..min(dE
[l],11*d-1)): od:

## SYSTEM FOR DIVISIBILITY BY y^(11*d)

Systemey11d := {}:

for k from 0 to (m-11)/3 do

Coeffs_y11d[11,m-11-3*k,k] := {coeffs(expand(mtaylor(Eq[11,m-11
-3*k,k], y, 11*d)), [x,y])}:

Systemey11d := Systemey11d union Coeffs_y11d[11,m-11-3*k,k]:

od:

Systeme_y11d := Systemey11d:

## VARIABLES FOR DIVISIBILITY BY y^(11*d)

Variables_y11d := indets(Systeme_y11d):

## RESOLUTION y^(11*d)

Solutions_y11d := solve(Systeme_y11d, Variables_y11d):

assign(Solutions_y11d):

##

for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
for l from 0 to m - 4*3 do nops(expand(EE||l)) od;

l := 0
1
l := 1
6
l := 2
6
l := 3
1

```

```

l := 4
l := 0
6
l := 1
l := 0
l := 0
l := 0

```

(12)

```

> ## EXPAND ALL UNKNOWNNS UP TO  $y^{(12*d)}$ 

for l from 0 to m - 0*3 do
AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j), j=0..min(dA
[l],12*d-1)): od:

for l from 0 to m - 1*3 do
BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j), j=0..min(dB
[l],12*d-1)): od:

for l from 0 to m - 2*3 do
CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j), j=0..min(dC
[l],12*d-1)): od:

for l from 0 to m - 3*3 do
DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j), j=0..min(dD
[l],12*d-1)): od:

for l from 0 to m - 4*3 do
EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j), j=0..min(dE
[l],12*d-1)): od:

## SYSTEM FOR DIVISIBILITY BY  $y^{(12*d)}$ 

Systemey12d := {}:

for k from 0 to (m-12)/3 do

Coeffs_y12d[12,m-12-3*k,k] := {coeffs(expand(mtaylor(Eq[12,m-12
-3*k,k], y, 12*d)), [x,y])}:

Systemey12d := Systemey12d union Coeffs_y12d[12,m-12-3*k,k]:

od:

Systeme_y12d := Systemey12d:

## VARIABLES FOR DIVISIBILITY BY  $y^{(12*d)}$ 

Variables_y12d := indets(Systeme_y12d):

## RESOLUTION  $y^{(12*d)}$ 

```

```
Solutions_y12d := solve(Systeme_y12d, Variables_y12d):
```

```
assign(Solutions_y12d):
```

```
##
```

```
for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
for l from 0 to m - 4*3 do nops(expand(EE||l)) od;
```

```
l := 0
```

```
1
```

```
l := 1
```

```
6
```

```
l := 2
```

```
6
```

```
l := 3
```

```
1
```

```
l := 4
```

```
l := 0
```

```
6
```

```
l := 1
```

```
l := 0
```

```
l := 0
```

```
l := 0
```

(13)

```
> ## EXPAND ALL UNKNOWNNS UP TO y^(13*d)
```

```
for l from 0 to m - 0*3 do
AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j), j=0..min(dA
[l],13*d-1)): od:
```

```
for l from 0 to m - 1*3 do
BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j), j=0..min(dB
[l],13*d-1)): od:
```

```
for l from 0 to m - 2*3 do
CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j), j=0..min(dC
[l],13*d-1)): od:
```

```
for l from 0 to m - 3*3 do
DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j), j=0..min(dD
[l],13*d-1)): od:
```

```

for l from 0 to m - 4*3 do
EE[l] := add(add(E[l][i,j]*x^i*y^j, i=0..dE[l]-j), j=0..min(dE
[l],13*d-1)): od:

## SYSTEM FOR DIVISIBILITY BY y^(13*d)

Systemey13d := {}:

for k from 0 to (m-13)/3 do

Coeffs_y13d[13,m-13-3*k,k] := {coeffs(expand(mtaylor(Eq[13,m-13
-3*k,k], y, 13*d)), [x,y])}:

Systemey13d := Systemey13d union Coeffs_y13d[13,m-13-3*k,k]:

od:

Systeme_y13d := Systemey13d:

## VARIABLES FOR DIVISIBILITY BY y^(13*d)

Variables_y13d := indets(Systeme_y13d):

## RESOLUTION y^(13*d)

Solutions_y13d := solve(Systeme_y13d, Variables_y13d):

assign(Solutions_y13d):

##

for l from 0 to m - 0*3 do nops(expand(AA[l])) od;
for l from 0 to m - 1*3 do nops(expand(BB[l])) od;
for l from 0 to m - 2*3 do nops(expand(CC[l])) od;
for l from 0 to m - 3*3 do nops(expand(DD[l])) od;
for l from 0 to m - 4*3 do nops(expand(EE[l])) od;

l := 0
1
l := 1
6
l := 2
6
l := 3
1
l := 4
l := 0
6
l := 1
l := 0

```

$l := 0$

$l := 0$

(14)

```
> ## EXPAND ALL UNKNOWNNS UP TO  $y^{(14*d)}$ 

for l from 0 to m - 0*3 do
AA||l := add(add(A||l[i,j]*x^i*y^j, i=0..dA[l]-j), j=0..min(dA
[l],14*d-1)): od:

for l from 0 to m - 1*3 do
BB||l := add(add(B||l[i,j]*x^i*y^j, i=0..dB[l]-j), j=0..min(dB
[l],14*d-1)): od:

for l from 0 to m - 2*3 do
CC||l := add(add(C||l[i,j]*x^i*y^j, i=0..dC[l]-j), j=0..min(dC
[l],14*d-1)): od:

for l from 0 to m - 3*3 do
DD||l := add(add(D||l[i,j]*x^i*y^j, i=0..dD[l]-j), j=0..min(dD
[l],14*d-1)): od:

for l from 0 to m - 4*3 do
EE||l := add(add(E||l[i,j]*x^i*y^j, i=0..dE[l]-j), j=0..min(dE
[l],14*d-1)): od:

## SYSTEM FOR DIVISIBILITY BY  $y^{(14*d)}$ 

Systemey14d := {}:

for k from 0 to (m-14)/3 do

Coeffs_y14d[14,m-14-3*k,k] := {coeffs(expand(mtaylor(Eq[14,m-14
-3*k,k], y, 14*d)), [x,y])}:

Systemey14d := Systemey14d union Coeffs_y14d[14,m-14-3*k,k]:

od:

Systeme_y14d := Systemey14d:

## VARIABLES FOR DIVISIBILITY BY  $y^{(14*d)}$ 

Variables_y14d := indets(Systeme_y14d):

## RESOLUTION  $y^{(14*d)}$ 

Solutions_y14d := solve(Systeme_y14d, Variables_y14d):

assign(Solutions_y14d):

##

for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
```

```

for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
for l from 0 to m - 4*3 do nops(expand(EE||l)) od;

```

```

l := 0

```

```

1

```

```

l := 1

```

```

6

```

```

l := 2

```

```

6

```

```

l := 3

```

```

1

```

```

l := 4

```

```

l := 0

```

```

6

```

```

l := 1

```

```

l := 0

```

```

l := 0

```

```

l := 0

```

(15)

```

> ## HOLOMORPHICITY AT INFINITY

```

```

for l from 0 to m - 0*3 do

```

```

  HA||l := add(add(A||l[i,j]*X^i*Y^j*T^(dA[l]-i-j), i=0..dA[l]-j),
j=0..dA[l]):

```

```

od:

```

```

#

```

```

for l from 0 to m - 1*3 do

```

```

  HB||l := add(add(B||l[i,j]*X^i*Y^j*T^(dB[l]-i-j), i=0..dB[l]-j),
j=0..dB[l]):

```

```

od:

```

```

#
for l from 0 to m - 2*3 do
  HC||l := add(add(C||l[i,j]*X^i*Y^j*T^(dC[l]-i-j), i=0..dC[l]-j),
j=0..dC[l]):
od:
#
for l from 0 to m - 3*3 do
  HD||l := add(add(D||l[i,j]*X^i*Y^j*T^(dD[l]-i-j), i=0..dD[l]-j),
j=0..dD[l]):
od:
#
for l from 0 to m - 4*3 do
  HE||l := add(add(E||l[i,j]*X^i*Y^j*T^(dE[l]-i-j), i=0..dE[l]-j),
j=0..dE[l]):
od:
##
for j from 0 to m - 0*3 do
  HK[j,0] := HA||j:
od:
#
for j from 0 to m - 1*3 do
  HK[j,1] := HB||j:
od:
#
for j from 0 to m - 2*3 do
  HK[j,2] := HC||j:
od:
#
for j from 0 to m - 3*3 do
  HK[j,3] := HD||j:

```



```

od:
#
for j from 0 to m - 4*3 do
HK[j,4] := HE||j:
od:

```

```

> ##
printlevel := 2:
for q from 0 to m/3 do
for p from 0 to m-3*q do

HEq[p,q] := expand(mtaylor(add(add(add(
  (-1)^k * 2^(p-r) * (d+1)^(j+k-p-q) * binomial(j,r) * (k!)/((p-
r)!*q!*(k+r-p-q)!))
* X^(m-q-j-2*k) * T^(k-q) * RY^(2*k-2*q) * HK[j,k],
j=r..m-3*k),
r=max(0,p+q-k)..min(p,m-3*k)),
k=q..m/3)
, T, m-p-3*q)):
od: od;

```

```

q := 0
HEq := table([ ])
HEq0,0 := 0
HEq1,0 := -TX4Y17AI2,17 - TX3Y18AI1,18 - TX2Y19AI0,19
HEq2,0 := 0
HEq3,0 := 0
q := 1
HEq0,1 := 0
q := 2

```

(16)

```

> HSysteme := {seq(seq(coeffs(HEq[p,q],[T,X,Y]),p=0..m-3*q),q=0..
m/3)} minus {0}:

HSolutions := solve(HSysteme):

assign(HSolutions):

##

for l from 0 to m - 0*3 do nops(expand(AA||l)) od;
for l from 0 to m - 1*3 do nops(expand(BB||l)) od;
for l from 0 to m - 2*3 do nops(expand(CC||l)) od;
for l from 0 to m - 3*3 do nops(expand(DD||l)) od;
for l from 0 to m - 4*3 do nops(expand(EE||l)) od;

l := 0
1
l := 1
3
l := 2
3
l := 3
1
l := 4
l := 0
3
l := 1
l := 0
l := 0
l := 0

```

(17)

```

> Jet_xR := factor(
+ (1/Ry^(m-0*2)) * x^(-twist) * add(AA||j * x1^(m-0*3-j) *
(R1/RR)^j * Delta_xR^0, j=0..m-0*3)
+ (1/Ry^(m-1*2)) * x^(-twist) * add(BB||j * x1^(m-1*3-j) *
(R1/RR)^j * Delta_xR^1, j=0..m-1*3)
+ (1/Ry^(m-2*2)) * x^(-twist) * add(CC||j * x1^(m-2*3-j) *
(R1/RR)^j * Delta_xR^2, j=0..m-2*3)
+ (1/Ry^(m-3*2)) * x^(-twist) * add(DD||j * x1^(m-3*3-j) *
(R1/RR)^j * Delta_xR^3, j=0..m-3*3)
+ (1/Ry^(m-4*2)) * x^(-twist) * add(EE||j * x1^(m-4*3-j) *
(R1/RR)^j * Delta_xR^4, j=0..m-4*3) );

```

$$Jet_xR := \frac{(R1 \, x \, x2 + 8 \, R1 \, x \, l^2 - R2 \, x \, x1) \, x^2 \, (AI_{1,17} \, x + AI_{0,18} \, y + AI_{0,17})}{5832 \, y^7 \, (x^9 + y^9 + 1)}$$

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$$\begin{aligned}
& \text{> Vars := indets(Jet_xR) minus \{x, x1, x2, y, R1, R2\};} \\
& \text{NbVars := nops(Vars);} \\
& \qquad \qquad \qquad \text{Vars := \{AI_{0,17}, AI_{0,18}, AI_{1,17}\}} \\
& \qquad \qquad \qquad \text{NbVars := 3}
\end{aligned} \tag{19}$$

$$\begin{aligned}
& \text{> Ens_Vars := \{\}:} \\
& \text{for i from 1 to nops(Vars) do} \\
& \text{Ens_Vars := Ens_Vars union \{op(i,Vars)=0\}:} \\
& \text{od;} \\
& \text{EnsVars := Ens_Vars;} \\
& \qquad \qquad \qquad \text{EnsVars := \{AI_{0,17}=0, AI_{0,18}=0, AI_{1,17}=0\}}
\end{aligned} \tag{20}$$

$$\begin{aligned}
& \text{> for i from 1 to nops(Vars) do} \\
& \text{JetSol[i] := factor(subs(EnsVars, subs(op(i,Vars)=1, Jet_xR))):} \\
& \text{od;} \\
& \qquad \qquad \qquad i := 1 \\
& \qquad \qquad \qquad \text{JetSol := table([])} \\
& \text{JetSol}_1 := \frac{(R1 \, x \, x2 + 8 \, R1 \, xI^2 - R2 \, x \, x1) \, x^2}{5832 \, y^7 \, (x^9 + y^9 + 1)} \\
& \qquad \qquad \qquad i := 2 \\
& \text{JetSol}_2 := \frac{(R1 \, x \, x2 + 8 \, R1 \, xI^2 - R2 \, x \, x1) \, x^2}{5832 \, y^6 \, (x^9 + y^9 + 1)} \\
& \qquad \qquad \qquad i := 3 \\
& \text{JetSol}_3 := \frac{(R1 \, x \, x2 + 8 \, R1 \, xI^2 - R2 \, x \, x1) \, x^3}{5832 \, y^7 \, (x^9 + y^9 + 1)} \\
& \qquad \qquad \qquad i := 4
\end{aligned} \tag{21}$$

